
BIOLOGY

9700/41

Paper 4 A Level Structured Questions

May/June 2017

MARK SCHEME

Maximum Mark: 100

Published

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Mark scheme abbreviations

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
<u>underline</u>	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore
AVP	alternative valid point

Question	Answer	Marks
1(a)	maintain / keep / restore, constant / stable / set-point / within narrow limits, internal environment / in body ;	1
1(b)(i)	A – pelvis ; <i>note if labelled medulla as affects ecf in part (ii)</i> B – ureter ;	2
1(b)(ii)	A full labels instead of letters <i>if region A (pelvis) was mislabelled as medulla in (i) can apply:</i> <i>ecf for L placed in pelvis</i> <i>ecf U placed in medulla only if word cortex also written by U/ultrafiltration</i> U – pointing to the cortex ; L – pointing to the medulla ; C – pointing to the renal vein ;	3

Question	Answer	Marks
1(c)	<i>max 5 of:</i> 1 <u>hypothalamus</u> detects (changes in) <u>water potential</u> (of the blood) ; 2 <u>osmoreceptors</u> shrink when, low / less, water in <u>blood</u> ; ora 3 ADH, produced / made, in hypothalamus ; 4 if low, water / Ψ, ADH secreted from <u>posterior pituitary</u> ; ora R ADH <i>produced</i> in posterior pituitary 5 <i>ref. to neurosecretory cells</i> or impulse / ADH transported, from hypothalamus to posterior pituitary ; 6 aquaporins ; 7 ADH increases <u>permeability</u> of, distal convoluted tubule / collecting duct ; ora 8 ADH causes, more water reabsorption / smaller volume of urine / more concentrated urine ; ora A both with and without ADH compared	5

Question	Answer	Marks
2(a)	1 high / increased / better / not limiting, carbon dioxide / temperature / (named) mineral ions ; I nutrients 2 (from) coral / polyp, respiration / metabolism or for algae, Calvin cycle / light independent reactions ; A correct use of mineral ions	2
2(b)	<i>max 4 of:</i> 1 (paper / thin layer) chromatography / chromatogram ; 2 place spot of, extract / pigments, on pencil mark / at base of, paper / TLC plate ; 3 dry and repeat (to concentrate spot) ; 4 dip, paper / chromatogram, in solvent / so solvent travels up paper ; A named organic solvent (I water) R if spot submerged 5 measure distance travelled by solvent (front) and pigment (spot) ; 6 (calculate) R_f value = $\frac{\text{distance travelled by pigment}}{\text{distance travelled by solvent (front)}}$; 7 look up / compare results with, known R_f values (to identify pigments) ;	4
2(c)	<i>max 3 of:</i> 1 pigments absorb, violet-blue / 400–490 nm / lamp colours, well / best / most / at 8 out of 10 peaks ; 2 <u>rate of photosynthesis</u> of algae increases with more light absorbed ; 3 coral growth (increases) with more (algal) photosynthesis ; R products respond to give growth 4 chlorophyll a and peridinin are, most abundant pigments / most important ; 5 AVP ; e.g. violet-blue / 400–490 nm, predominate at the depths where corals live	3

Question	Answer	Marks
3(a)	<i>max 2 of:</i> 1 can, kill / control, weeds ; R kill, insects / pests 2 reduce competition / increase yield (of rape) ; 3 AVP ; e.g. manual weeding / hoeing, difficult / expensive <i>ref. to</i> glufosinate converted to non-toxic compound	2
3(b)(i)	circle of / circular, DNA ; I loop R single-stranded small / supplementary ;	2
3(b)(ii)	<i>max 3 of:</i> 1 small so can be taken up by, cells / bacteria ; 2 replicate, independently / fast ; A have <i>ori</i> / origin of replication / high copy number 3 (DNA) has restriction site(s) / can be cut by restriction enzymes ; A have polylinker 4 have, marker genes / genes for resistance (for screening) ; 5 AVP ; e.g. circular so, increased stability / reduced host cell degradation	3
3(b)(iii)	<i>max 2 of:</i> 1 <u>RNA polymerase</u> binds ; 2 so, transcription / mRNA synthesis, begins / occurs / allowed ; 3 AVP ; e.g. correct / template, strand is transcribed <i>ref. to</i> tissue-specific / inducible, expression	2
3(c)(i)	28 ;	1

Question	Answer	Marks
3(c)(ii)	<i>max 1 of:</i> spray with herbicide and, those that die did not have the <i>bar</i> gene / those that survive did have the <i>bar</i> gene ; add gene for fluorescence with <i>bar</i> gene and test plants under UV / use PCR with primer complementary to <i>bar</i> gene / use (gene) probe (on Southern blot) of electrophoresis gel ;	1
3(c)(iii)	<i>max 3 of:</i> <i>advantage of male sterile GM variety</i> 1 avoid transferring, <i>bar</i> / resistance, gene to wild, radish / relations ; ora 2 avoid superweeds ; ora 3 avoid <u>type 2</u> hybrids ; ora <i>disadvantage of type 2 hybrids (from GM variety that produces pollen)</i> 4 taller (than wild radish) ; A very tall / 88 cm / 95 cm 5 produce, more / many, seeds (than wild radish) ; A 3958 / 443 more 6 may (out)compete, wild radish / crops ;	3

Question	Answer	Marks															
4(a)	<table border="1"> <thead> <tr> <th>stage of respiration</th><th>ATP used</th><th>ATP produced</th></tr> </thead> <tbody> <tr> <td>glycolysis</td><td>yes</td><td>yes</td></tr> <tr> <td>link reaction</td><td>no</td><td>no</td></tr> <tr> <td>Krebs cycle</td><td>no</td><td>yes</td></tr> <tr> <td>oxidative phosphorylation</td><td>no</td><td>yes</td></tr> </tbody> </table> ;; <i>4 correct = 2 marks, 2 or 3 rows correct = 1 mark</i> <i>If ticks and crosses used need all 4 correct for maximum 1 mark</i>	stage of respiration	ATP used	ATP produced	glycolysis	yes	yes	link reaction	no	no	Krebs cycle	no	yes	oxidative phosphorylation	no	yes	2
stage of respiration	ATP used	ATP produced															
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Question	Answer	Marks
4(b)	<i>max 5 of:</i> <i>group A (accept ora for group B throughout) accept 'they' = group A</i> 1 higher <u>ratio</u> ; 2 larger / more, inner membrane / cristae (than B) ; 3 more, ETCs / cytochromes / ATP synth(et)ase / stalked particles ; I ATPase 4 oxidative phosphorylation ; 5 more ATP produced ; 6 <u>muscles</u> can contract for, longer / more time / without getting tired ; I exercise longer I muscles contract faster 7 AVP ; e.g. chemiosmosis or detail thereof: H⁺ move, down gradient / through ATP synth(et)ase I ATPase <i>If B and A switched round penalise once only</i>	5

Question	Answer	Marks
5(a)	A all figures $\pm 1\%$ I increase or decrease 'by x%' when difference from start time to end calculated max 4 of: 1 decrease in daisies and normal poppies, overall / in 6 years / after 1999–2000 ; 2 (decrease in, daisies / normal poppies) from 50% to 15% ; 3 increase in poppy biotype X from 1% to 70% ; 4 increase in, total / combined, poppies from 52% to 85% ; 5 daisies and normal red poppies are always equal in % frequency ; A remain equal 6 steep / huge / dramatic, decrease in daisies and normal red poppy after 2001 or increase in X, is steeper after 2001 ;	4
5(b)(i)	max 3 of: 1 change in <u>primary</u> structure ; 2 change in, tertiary / 3D / globular, structure ; 3 <u>active site</u>, binds substrate / forms ESC ; 4 metsulfuron-methyl does not, inhibit / bind to, enzyme ; 5 enzyme, functions / forms amino acids ;	3

Question	Answer	Marks
5(b)(iii)	<i>max 3 of:</i> <i>in 2003 (compared to 1998)</i> 1 more, weeds / poppies, and less wheat / higher proportion of weeds in wheat ; I wheat yield 2 most weeds are now, poppy biotype X / resistant to herbicide ; 3 poppy biotype X, not killed by / resistant to herbicide ; 4 <u>wheat</u> have, more competition for / less access to, space / light / water / minerals ; I nutrients	3
5(c)	<i>max 3 of:</i> 1 (biotype X) poppies, die / do not survive / do not breed ; 2 their, numbers / abundance. would decrease ; 3 selection pressure, removed / changed / new ; 4 (biotype X) mutant / resistance, <u>allele</u> no longer, advantageous / selected for / passed on ; 5 possibility of beneficial mutation in gene for different enzyme or could adapt / evolve resistance, to new herbicide ; R if new herbicide causes mutation	3

Question	Answer	Marks
6(a)(i)	<i>max 2 of:</i> 1 tropomyosin / it, covers / uncovers, myosin binding sites on actin ; R inhibits R active site 2 when calcium ions bind to troponin, tropomyosin / it, moves / changes shape ; 3 allows myosin to, bind to actin / form cross-bridges ; ora	2

Question	Answer	Marks
6(a)(ii)	<i>max 4 of:</i> 1 ATP hydrolysis / $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$; 2 (causes myosin) <u>head</u> to, pivot / rotate / tilt / stand up ; 3 myosin / head, binds to actin / forms cross-bridges with actin ; R active site 4 ADP and P_i detach ; 5 (myosin) <u>head</u>, swings back / returns to previous position ; 6 actin is moved / power stroke occurs ; 7 (new) ATP binds ; 8 myosin / head, detaches from actin / cross-bridges break ; <i>A mps in any order apart from 1, 4 and 7 which must be linked to correct action</i>	4
6(b)(i)	<i>max 2 of:</i> 1 to, supply / provide, (enough / plenty of) <u>glucose</u> ; 2 for <u>glycolysis</u> ; 3 as little ATP is produced by anaerobic respiration ; 4 as few capillaries are present (to supply glucose directly) ;	2
6(b)(ii)	<i>max 2 of:</i> 1 to, supply / provide, (enough / plenty of) <u>oxygen</u> ; 2 aerobic respiration / oxidative phosphorylation ; 3 to remove, carbon dioxide / lactate ; A lactic acid 4 to, avoid fatigue or promote, stamina / endurance (for exercise / work) ;	2

Question	Answer	Marks
8(a)	<i>max 4 of:</i> 1 different <u>habitats</u> ; 2 different <u>niches</u> ; 3 many (different) <u>species</u> / large variety of <u>species</u> ; 4 <i>ref. to</i> (much) genetic diversity within a species ; 5 different <u>selection</u> pressures ; 6 <i>ref. to</i> <u>adaptation</u> ; 7 different, climate / rainfall / temperature / soil / topography / conditions ;	4
8(b)(i)	both sites are the same / no (significant) difference between two sites ;	1
8(b)(ii)	<i>genera 2 and species 4 ;</i>	1

Question	Answer	Marks																								
8(b)(iii)	<i>all figures to 3 d.p. to score but only penalise extra d.p. or rounding error associated with extra d.p. once</i> <table><tr><th>species</th><th>number on grassland not grazed</th><th>n / N</th><th>$(n / N)^2$</th></tr><tr><td><i>Onthophagus pennsylvanicus</i></td><td>6641</td><td><u>0.873</u></td><td>0.762</td></tr><tr><td><i>Canthon ebenus</i></td><td>774</td><td><u>0.102</u></td><td>0.010</td></tr><tr><td><i>Canthon pilularius</i></td><td>108</td><td><u>0.014</u></td><td>0.000</td></tr><tr><td><i>Onthophagus hecate</i></td><td>85</td><td><u>0.011</u></td><td>0.000</td></tr><tr><td>total</td><td>7608</td><td></td><td>0.772</td></tr></table> n / N figures correct / numbers of each species divided by total ; $(n / N)^2$ calculated and added up ; ecf from incorrect column 1 including figures with fewer / more than 3 d.p. 0.228 ; ecf total figure subtracted from 1	species	number on grassland not grazed	n / N	$(n / N)^2$	<i>Onthophagus pennsylvanicus</i>	6641	<u>0.873</u>	0.762	<i>Canthon ebenus</i>	774	<u>0.102</u>	0.010	<i>Canthon pilularius</i>	108	<u>0.014</u>	0.000	<i>Onthophagus hecate</i>	85	<u>0.011</u>	0.000	total	7608		0.772	3
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8(b)(iv)	greater species <u>evenness</u> on grazed grassland ; ora A mostly, one species / <i>O. pennsylvanicus</i>, on not grazed grazing increases (dung beetle species) (bio)<u>diversity</u> ; ora <i>if opposite conclusion reached check answer for (iii) and apply ecf for mp2 if $D > 0.521$</i>	2																								

Question	Answer	Marks
9(a)	<i>max 8 of:</i> 1 stick has, pad containing / immobilised, enzymes ; 2 glucose oxidase ; 3 peroxidase ; 4 stick dipped in urine ; A person, urinates / AW, on stick 5 glucose reacts to give hydrogen peroxide ; 6 (hydrogen peroxide reacts with) colourless substance / chromogen ; R dye / pigment 7 to give, colour change / coloured substance ; A change to any named colour 8 compare with colour chart ; 9 more glucose gives darker colour ; 10 specific / only detects glucose ; 11 AVP ; e.g. does not give current <u>blood</u> glucose concentration not numerical	8

Question	Answer	Marks
9(b)	<i>max 7 of:</i> <i>high blood glucose concentration</i> 1 detected by β cells ; I alpha cells I receptors 2 in, islets of Langerhans / pancreas ; 3 (more) insulin secreted ; I produced 4 into blood ; 5 increases glucose absorption in liver (by phosphorylating glucose) ; 6 increases permeability to glucose in, muscle / fat, cells or adds GLUT 4 proteins to cell surface membranes of, muscle / fat, cells ; 7 increases (rate of) respiration of <u>glucose</u> ; 8 conversion of glucose to <u>glycogen</u> / <u>glycogenesis</u> ; 9 inhibits secretion of <u>glucagon</u> / decreases <u>gluconeogenesis</u> ; 10 <u>negative feedback</u> ;	7

Question	Answer	Marks
10(a)	<i>max 6 of:</i> 1 dendrites (lead to cell body) ; R at both ends 2 nucleus in, cell body / soma ; R if cell body not at one end 3 many mitochondria (in cell body) ; 4 much RER / Nissl's granules (in cell body) ; 5 long / one, <u>axon</u> ; A an <u>axon</u> 6 synaptic, knobs / termini / boutons, at end furthest from cell body ; 7 Schwann cells / myelin ; 8 nodes of Ranvier ; <i>accept points on labelled diagram</i>	6

Question	Answer	Marks
10(b)	<i>max 9 of:</i> 1 Na^+ / sodium ion, channels open ; I ligand or voltage gated 2 Na^+ enters, cell / axon ; A Na ions / sodium ions 3 inside / p.d., becomes, less negative / positive / +40 mV or causes depolarisation (in correct context) ; 4 Na^+ / sodium ion, channels close ; ecf from mp1 I ligand or voltage-gated 5 K^+ / potassium ion, channels open ; ecf from mp1 I ligand or voltage-gated 6 K^+ moves out (of cell) ; A K ions / potassium ions 7 inside / p.d., becomes negative / A negative figure or causes repolarisation (in correct context) ; 8 <u>local circuits</u> ; 9 myelin (sheath) / Schwann cells, insulate / prevent ion movement ; 10 action potential / depolarisation, <u>only</u> at, nodes (of Ranvier) ; 11 saltatory conduction / action potential jumps from node to node ; A impulse for AP 12 one-way / unidirectional, transmission ; 13 AVP ; e.g. hyperpolarisation / refractory period	9